

Flow of control

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You need to buy a pen!

Mother paid £100 for pen

if cost_of_pen $\leq$ 100 ?

else:

print("The pen is in our budget")

print("The pen is not in my budget")

$\leq$ ↑ Maths

$< =$ ↑ computer

Selection: A decision involves selecting from one of the two or more possible options. In programming, this concept of decision making or selection is implemented with the help of **if...else** statement.

The syntax is:

```
if <condition>:
```

```
    Statement(s)
```

```
else: #optional
```

```
    Statement(s)
```

The if else statement allow us to write two alternative paths and the control, Condition determines which path gets executed.

"elif" statement: If we have more selection then we can use elif statement:

Syntax:

```
if <condition>:
```

```
    Statement(s)
```

```
elif <condition>: #optional
```

```
    Statement(s)
```

```
elif <condition>: #optional
```

```
    Statement(s)
```

```
elif <condition>: #optional
```

```
    Statement(s)
```

```
...
```

```
else:
```

```
    Statement(s)
```

Write a program in Python to take input of a number. And, check whether a number is positive, negative or 0.

```
num = int(input('Enter number: '))
if num == 0:
    print('Zero')
elif num > 0:
    print('positive')
else:
    print('negative')
```

Output:

```
PS C:\Users\bluej\Doc
Students/ShivRay/ForG
Enter number: 67
positive
PS C:\Users\bluej\Doc
Students/ShivRay/ForG
Enter number: 0
Zero
PS C:\Users\bluej\Doc
Students/ShivRay/ForG
Enter number: -1
negative
```

Write a program to display the appropriate message as per the color of the signal at the road crossing.

Input: red or RED

Output: stop

Input: orange or ORANGE

Output: slow

Input: green or Green

Output: go

A library charges a fine for books returned late as given below:

First five days ₹1 per day

Six to ten days ₹2 per day

Above ten days ₹3 per day

Design a program in Python (with relevant supporting material) to calculate the fine assuming that a book is returned N days late.

Home work:

☒ Understanding the Problem (Concept)

This is a **piecewise (slab-based)** calculation problem. The fine depends on ranges of days:

- **1-5 days** → ₹1/day
- **6-10 days** → ₹2/day
- **Above 10 days** → ₹3/day

This means the fine is **cumulative**, not flat. For example:

If a book is 12 days late:

- First 5 days → $5 \times 1 = ₹5$
- Next 5 days → $5 \times 2 = ₹10$
- Remaining 2 days → $2 \times 3 = ₹6$
- **Total = ₹21**

Python Program

```
# Program to calculate library fine
# Input number of late days
n = int(input("Enter number of days late: "))
fine = 0
if n <= 5:
    fine = n * 1
elif n <= 10:
    fine = (5 * 1) + (n - 5) * 2
else:
    fine = (5 * 1) + (5 * 2) + (n - 10) * 3
print("Total fine = ₹", fine)
```

🔗 Key Concepts Used

- Conditional statements (if-elif-else)
- Arithmetic operations
- Piecewise logic (slab system)
- Input/Output handling

🔍 Alternative Approach (Loop-based Logic)

```
n = int(input("Enter number of days late: "))
fine = 0
for day in range(1, n + 1):
    if day <= 5:
        fine += 1
    elif day <= 10:
        fine += 2
    else:
        fine += 3
print("Total fine = ₹", fine)
```

🔗 This approach is useful for understanding **iteration + conditional flow**

Home work

1. Electricity Bill Calculator

Design a program to calculate electricity bill:

- First 100 units → ₹1/unit
 - Next 100 units → ₹2/unit
 - Above 200 → ₹3/unit
- Include meter charge ₹50.

2. Student Grade System

Input marks of 5 subjects and:

- Calculate percentage
- Assign grade:
 - ≥ 90 → A
 - 75-89 → B
 - 50-74 → C
 - < 50 → Fail

3. Bank Interest Calculator

Calculate compound interest based on:

- Principal, Rate, Time
- Show yearly breakdown

4. Shopping Discount System

Apply discount:

- ₹0-1000 → 5%
 - ₹1001-5000 → 10%
 - Above ₹5000 → 15%
- Generate final bill.

5. Parking Fee System

Charges:

- First 2 hours → ₹10/hour
- Next 3 hours → ₹20/hour

- After that → ₹30/hour

6. Telephone Bill Program

- First 100 calls → ₹1/call
- Next 100 → ₹1.5/call
- Above 200 → ₹2/call
- Fixed monthly charge ₹200

7. Income Tax Calculator

Based on slabs:

- Up to ₹2.5L → No tax
- ₹2.5L–5L → 5%
- ₹5L–10L → 20%
- Above ₹10L → 30%

8. Movie Ticket Booking System

- Age-based pricing:
 - <12 → ₹100
 - 12–60 → ₹200
 - 60 → ₹150
- Add weekend surcharge ₹50.

9. Courier Charge Calculator

Weight-based charges:

- First 2 kg → ₹50/kg
- Next 3 kg → ₹40/kg
- Above → ₹30/kg

10. Attendance & Fine System

If attendance <75%:

- 70–74 → ₹100 fine
- 60–69 → ₹200 fine
- Below 60 → ₹500 fine + warning